



CS 201P: Computer Organization Laboratory

Course Code	CS 201P
Course Name	Computer Organization Laboratory
L-T-P-C	0-0-2-1
Prerequisites	None
Intended for	UG
Distribution	Discipline Core
Semester	V
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

Lab assignments (listed below) require 3 hours in the lab, preceded by at least 3 hours at home. The weekly assignments would be targeted at understanding the concepts covered in the theory course. Some examples of the structure of assignment is as follows:

- Week 1 Getting familiarity in the Verilog/VHDL programming.
- Week 2-3 Design and simulation of simple combinational and sequential circuits (flip flops) using Verilog/VHDL programming.
- Week 4 Design and simulation of 16-bit signed and unsigned integer adder-subtractor circuit.
- Week 5 Design and simulation of 16-bit signed and unsigned integer multiplication circuit using carry save addition.
- Week 6 Design and simulation of 16-bit signed and unsigned integer combinational division circuit using non-restoring procedure.
- Week 7 Design and simulation of 16-bit signed and unsigned arithmetic and logic unit (ALU) as a single unit by combining all the circuits simulated from week 2-7.
- Week 8-13 Programming in assembly language, e.g., assembly language in IA32 architecture

Text Books

1. Sivarama P. Dandamudi, **Guide to Assembly Language Programming in Linux**, Springer, 2005.